

Lab 4: Liquid Chemistry

BIOL-1

Name: _____ Date: _____

Learning Objectives

1. **Define diffusion** and explain how medium density/viscosity affects the rate of diffusion.
2. **Observe viscosity** directly by comparing fluid motion in water versus methyl cellulose.
3. **Demonstrate redox reactions** using a common oxidizer (bleach) and an indicator dye.
4. **Perform serial dilutions** to create a concentration gradient and understand how visual color relates to solute amount.

Purpose

Perform simplified experiments to observe diffusion, viscosity, redox reactions, and serial dilutions using minimal reagents.

Connections to Biology

- **Diffusion** is the primary method for moving small molecules (oxygen, carbon dioxide) in and out of cells.
 - **Viscosity** of bodily fluids (blood plasma, cytoplasm) affects how easily cells and molecules move through them.
 - **Redox reactions** are fundamental to metabolism (cellular respiration and photosynthesis).
 - **Concentration gradients** drive many biological processes, including nerve impulses and nutrient absorption.
 - **Organic molecules tie-in:** Methyl cellulose is a **carbohydrate polymer** (a cellulose derivative). Its thickness shows how the size and structure of organic macromolecules change the physical behavior of a solution — the same reason a cell's cytoplasm is viscous.
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1. Diffusion and Viscosity (Water vs. Methyl Cellulose)

Goal: Compare properly how red dye diffuses and moves in standard water versus a thick, viscous solution (simulating cytoplasm or mucus).

Materials

- 2 test tubes
- Distilled water
- Methyl cellulose solution
- Red food color

Procedure

1. **Fill tube A:** $\frac{3}{4}$ full with distilled water.
2. **Fill tube B:** Same volume ($\frac{3}{4}$ full) of methyl cellulose solution.
3. Use a straw or dropper to place **one** small drop of red food color at the surface of **Tube A**.
Do not stir.
4. Place the same size drop at the surface of **Tube B**.
5. Watch for 5–15 minutes. Note how quickly the dye spreads (diffuses/mixes) and if it sinks or stays suspended.

Questions

1. In which tube did the dye spread out faster? Why?

2. How would you define **viscosity** based on what you see in Tube B?

3. Imagine a cell needs to move oxygen quickly from its membrane to its center. Would this happen faster if the cytoplasm were like water or like methyl cellulose?

2. Redox Reaction (Methylene Blue + Bleach)

Goal: Show that an oxidizer (bleach) chemically alters and destroys the color of methylene blue.

Safety

- Work with drops, not large volumes.

Materials

- 1 test tube
- Methylene blue solution
- 10% bleach
- Distilled water

Procedure

1. Fill the test tube $\frac{1}{2}$ full with distilled water.
2. Add methylene blue dropwise until the water is a clear, medium blue.
3. Add **1 drop** of bleach and gently swirl.
4. Continue adding bleach one drop at a time, swirling after each addition. Count how many drops it takes until the solution becomes nearly colorless.

Questions

1. What happened to the blue color? Did the dye leave the tube, or did it change form?

2. Bleach is an **oxidizing agent**. What does it do to the electrons in the methylene blue molecule?

3. Potassium Permanganate Serial Dilution

Goal: Create a visual concentration gradient to understand how "dilution" works.

Safety

- Work with drops, not large volumes.

Materials

- 2 test tubes
- Potassium permanganate stock solution
- Distilled water

Procedure

1. **Tube 1 (Stock):** Add 5 mL of the **Potassium Permanganate Stock** solution to **Tube 1**.
2. **Tube 2 (Dilution):** Add 5 mL of clear water to **Tube 2**.
3. **Transfer:** Take 1 mL (or a pipette squeeze) from **Tube 1** and add it to **Tube 2**. Mix gently.

Questions

1. Which tube has the highest concentration of potassium permanganate? Which has the lowest?

2. Even in the lighter tube, can you still see some pink/purple color? What does this tell you about the molecules in the solution?

3. If you took a drop from Tube 2 and put it under a microscope, would you expect to see solid crystals or dissolved ions?